

Day 9 — Routing Fundamentals & Routing Tables

Now that you understand **IP addressing, subnet masks, same/different networks**, we can move to the next topic in our schedule:

How does a router actually decide where to send an IP packet?

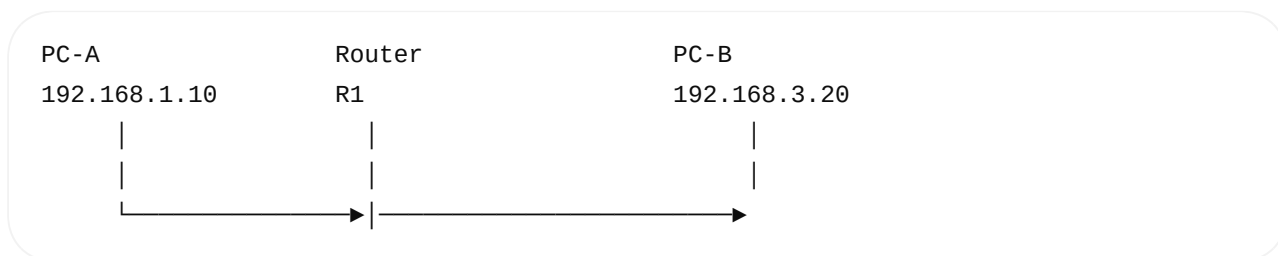
This is the main question of Day 9.

1. First: What is Routing?

Routing means:

Choosing a path for an IP packet to reach its destination network.

Example:



PC-A wants to send data to:

192.168.3.20

PC-A knows:

192.168.3.20

is **not in its local network**.

So it sends the packet to its:

Default Gateway

The router receives it and asks:

"Where should I send this packet next?"

That decision is **routing**.

2. Routing Happens at Layer 3

Remember our OSI model:

Layer 7 Application
Layer 6 Presentation
Layer 5 Session
Layer 4 Transport TCP / UDP
Layer 3 Network ← ROUTING
Layer 2 Data Link Ethernet / MAC
Layer 1 Physical

The router's main forwarding decision is a **Layer 3 / IP decision**.

The router looks primarily at:

Destination IP

and uses its:

Routing Table

3. What is a Routing Table?

A routing table is basically a list of:

Destination Network
+
Where to send it

For example:

Destination	Next Hop	Interface

192.168.1.0/24	directly	eth0
192.168.2.0/24	10.0.0.2	eth1
192.168.3.0/24	10.0.0.3	eth1
0.0.0.0/0	10.0.0.1	eth1

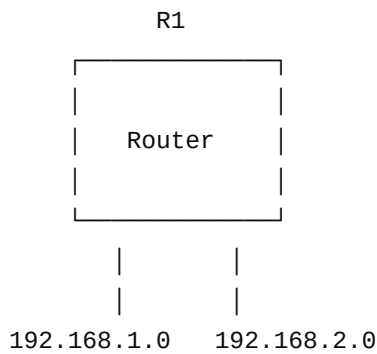
Don't worry about every column yet.

The important idea is:

Destination IP
↓
Routing Table
↓
Select route
↓
Forward packet

4. Simple Example

Suppose router R1 has:



Its routing table might contain:

```
192.168.1.0/24 → eth0
192.168.2.0/24 → eth1
```

Now packet arrives:

```
Destination IP:
192.168.2.50
```

Router checks:

```
192.168.2.50
  ↓
Matches 192.168.2.0/24
  ↓
Use eth1
```

Then:

```
Router
  |
  | eth1
  ▼
192.168.2.50
```

5. The Router Does NOT Ask "Which Exact IP?"

This is important.

The routing table usually contains **networks/prefixes**, not one entry for every host.

Example:

192.168.2.0/24

means:

192.168.2.0

↓

192.168.2.1

↓

...

↓

192.168.2.254

So one route can cover an entire network.

6. Network Prefix

Consider:

192.168.2.0/24

The router interprets:

192.168.2

|

└─ Network prefix

The host portion is:

last 8 bits

Therefore the route:

192.168.2.0/24

matches destinations such as:

192.168.2.10

192.168.2.50

192.168.2.100

192.168.2.200

7. Directly Connected Route

Suppose router has:

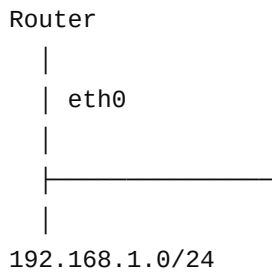
eth0 = 192.168.1.1/24

The router automatically knows:

192.168.1.0/24

is directly connected to eth0 .

Conceptually:



Routing table:

192.168.1.0/24 → directly connected → eth0

So if destination is:

192.168.1.50

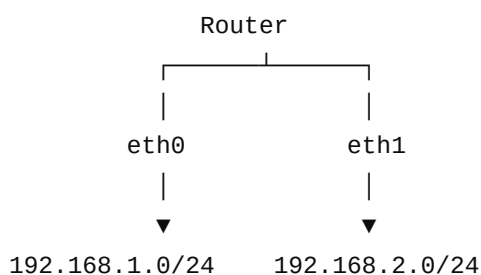
the router doesn't need another router.

8. What Does "Directly Connected" Mean?

It means:

The destination network is attached directly to one of the router's interfaces.

Example:



The router directly knows both networks.

9. Next Hop

Now consider:



PC-B belongs to:

192.168.3.0/24

R1 doesn't have that network directly connected.

Its routing table might say:

192.168.3.0/24 → 10.0.0.2

Here:

10.0.0.2

is the **next-hop router**.

So:

Destination:

192.168.3.20

Routing table:

192.168.3.0/24

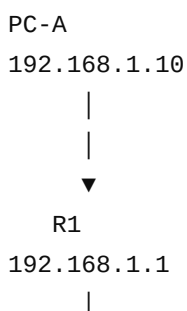
↓

Next hop = 10.0.0.2

R1 sends the packet toward R2.

10. Full Routing Example

Let's build a complete network.





PC-A wants:

192.168.3.20

11. Step 1 — PC-A Checks the Destination

PC-A:

192.168.1.10/24

Destination:

192.168.3.20

PC-A calculates:

My network:
192.168.1.0/24

Destination network:
192.168.3.0/24

Different.

Therefore:

Send to default gateway

12. Step 2 — PC-A Uses ARP

PC-A needs the MAC address of its gateway:

192.168.1.1

So it performs ARP:

```
PC-A
  |
  | Who has 192.168.1.1?
  ▼
Broadcast
  |
  ▼
Router R1
  |
  | I have it
  ▼
MAC-R1
```

Now PC-A knows:

```
Gateway IP:
192.168.1.1

Gateway MAC:
AA:AA:AA:AA:AA:AA
```

13. Step 3 — PC-A Creates the Packet

Layer 3:

```
| Source IP
| 192.168.1.10
|
| Destination IP
| 192.168.3.20
|
```

Notice:

Destination IP is PC-B.

Not the router.

14. Step 4 — Ethernet Frame

But the Layer 2 destination is:

```
Router's MAC
```

So:

Ethernet
Source MAC = PC-A
Destination MAC = R1
IP Packet
Source IP = 192.168.1.10
Destination IP = 192.168.3.20

This is one of the most important concepts from Days 7–9.

15. Step 5 — Router R1 Receives the Frame

R1 receives the Ethernet frame.

First:

Layer 2
↓
Ethernet destination MAC
↓
Is this frame for me?
↓
YES

R1 removes the Ethernet header/trailer and examines the IP packet.

16. Step 6 — R1 Looks at Destination IP

R1 sees:

Destination IP:
192.168.3.20

It checks its routing table.

Example:

Destination	Next Hop

192.168.1.0/24	Direct
10.0.0.0/30	Direct
192.168.3.0/24	10.0.0.2

R1 finds:

192.168.3.0/24

Therefore:

Next hop:
10.0.0.2

17. Step 7 — R1 Creates a New Ethernet Frame

This is extremely important.

The IP packet is forwarded, but the **Layer 2 frame is rebuilt for the next link**.

Before R1:

PC-A → R1

MAC:

PC-A → R1

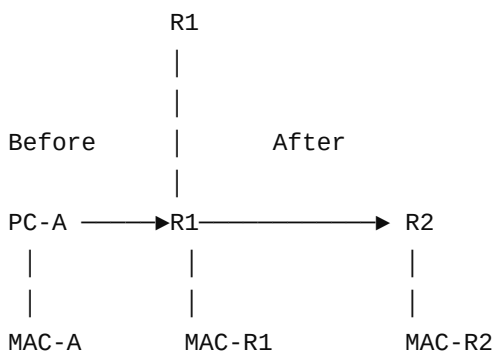
After R1:

R1 → R2

MAC:

R1 → R2

Visual:



More precisely:

Link 1:

Src MAC = PC-A
Dst MAC = R1



Link 2:

Src MAC = R1

Dst MAC = R2

18. But What Happens to IP?

The destination IP remains:

192.168.3.20

Conceptually:

First link:

IP:

192.168.1.10 → 192.168.3.20

MAC:

PC-A → R1

Second link:

IP:

192.168.1.10 → 192.168.3.20

MAC:

R1 → R2

Therefore:

IP = end-to-end destination

MAC = hop-to-hop delivery

★ **Memorize this.**

19. Step 8 — R2 Receives the Packet

R2 receives the frame.

It examines:

Destination IP:

192.168.3.20

R2 has:

192.168.3.0/24

directly connected.

Therefore:

192.168.3.20

↓

Directly connected network

↓

Send toward PC-B

20. Step 9 — R2 Needs PC-B's MAC

R2 needs the MAC address associated with:

192.168.3.20

It can use ARP on that local network.

Conceptually:

R2

|

| ARP:

| Who has 192.168.3.20?

▼

Network

|

▼

PC-B

|

| My MAC is BB:BB:...

▼

R2

21. Step 10 — Final Frame

R2 creates the final Ethernet frame:

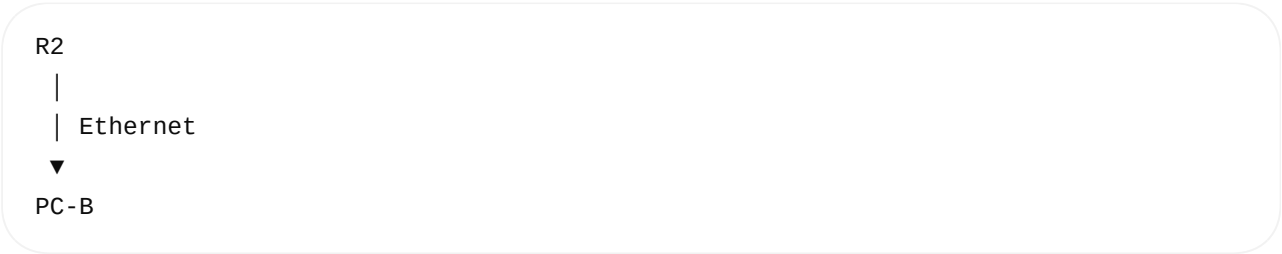
Ethernet

Source MAC = R2

Destination MAC = PC-B

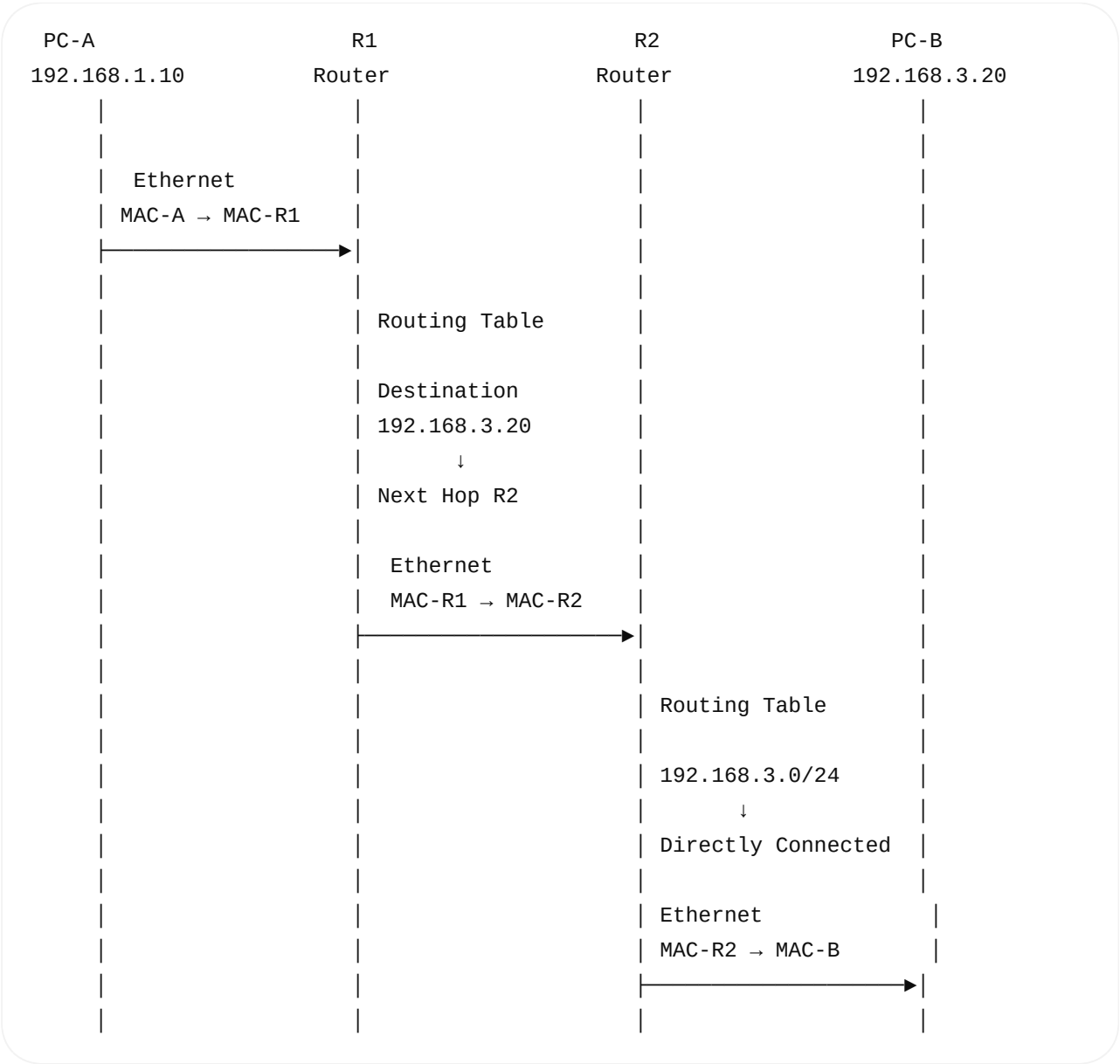


Then:



22. Complete Visualization

This is the diagram I want you to understand:



23. The Three Different Decisions

This is the easiest way to organize everything you've learned.

PC-A first asks:

Is PC-B in my network?

Uses:

IP + Subnet Mask

If different network:

Which router should I send to?

Uses:

Default Gateway

Router asks:

Where should I forward this packet?

Uses:

Destination IP
↓
Routing Table
↓
Longest Prefix Match
↓
Next Hop / Interface

24. Longest Prefix Match

Now let's connect this to the subnetting from Day 8.

Suppose router has:

10.0.0.0/8
10.10.0.0/16
10.10.20.0/24
0.0.0.0/0

Destination:

10.10.20.50

All of these can potentially match:

10.0.0.0/8
↓
10.10.0.0/16
↓
10.10.20.0/24

Which one does the router choose?

/24

because it is the **most specific route**.

So:

Larger prefix length
↓
More specific
↓
Higher priority

25. What is the Default Route?

You will frequently see:

0.0.0.0/0

This means:

"If nothing more specific matches, use this route."

Example:

Routing table

192.168.1.0/24 → eth0

Destination:

8.8.8.8

Does it match:

192.168.1.0/24 ? NO

192.168.2.0/24 ? NO

Then:

0.0.0.0/0

↓

Default route

26. Linux Routing Table

On your Ubuntu system, run:

```
ip route
```

You may see something like:

```
default via 192.168.1.1 dev wlp2s0
```

```
192.168.1.0/24 dev wlp2s0 proto kernel scope link src 192.168.1.20
```

Interpretation:

First line

```
default via 192.168.1.1
```

means:

```
Unknown destination
```

↓

```
Send to gateway
```

```
192.168.1.1
```

Second line

```
192.168.1.0/24 dev wlp2s0
```

means:

192.168.1.0/24



Directly connected



Use wlp2s0

27. Very Important Linux Command

You can ask Linux:

"How would you route this IP?"

Use:

```
ip route get 8.8.8.8
```

For example, conceptually:

8.8.8.8



Routing table



Default route



Gateway



Interface

This is a very useful troubleshooting command.

28. Router vs Switch

You should now clearly distinguish them.

Switch

Mainly Layer 2

Uses MAC addresses

Uses MAC address table

Forwards Ethernet frames

Router

Mainly Layer 3

Uses IP addresses

Uses routing table

Forwards IP packets

Switch

Connects devices within LAN/VLAN

Learns MAC addresses

Router

Connects different IP networks

Learns/uses routes

Visual:

Switch:

MAC Address



MAC Table



Ethernet Frame



Port

Router:

Destination IP



Routing Table



Next Hop / Interface



New Layer-2 Frame

29. One Important Correction From Your Earlier Learning

You previously asked whether:

"Router only does rework in Layer 2."

The better way to understand it is:

Router makes its forwarding decision at Layer 3, but it also has to process Layer 2 on each attached link.

For example:

Incoming:

Ethernet



IP



Routing decision



New Ethernet



Outgoing

So the router does **not** simply "work at Layer 2."

It:

Receives Layer-2 frame



Processes Layer-3 IP packet



Makes routing decision

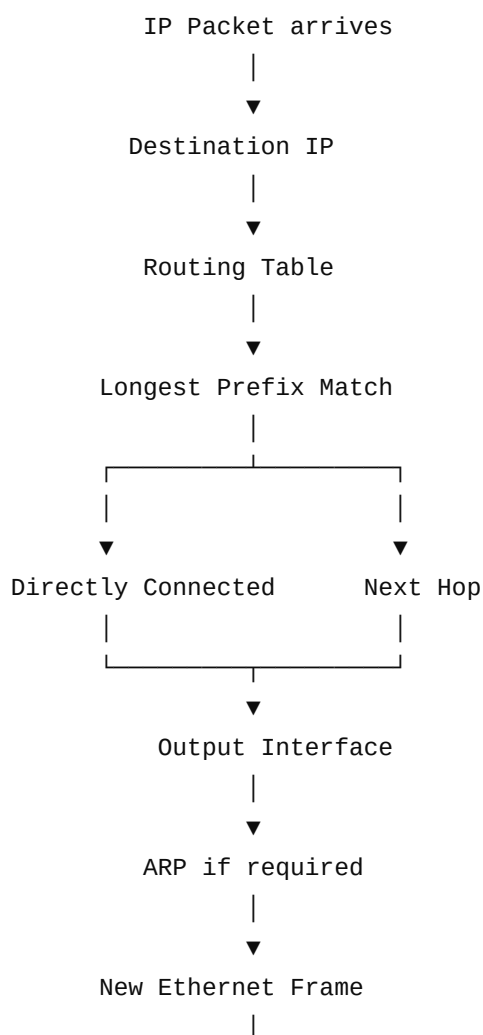


Creates outgoing Layer-2 frame

That's the exact model you should use in interviews.

30. ★ Day 9 Core Concept

Memorize this flow:



And remember this sentence for your interview:

A router examines the destination IP address, looks up the best matching route in its routing table using longest-prefix matching, selects an outgoing interface/next hop, and forwards the packet using a new Layer-2 frame on the next link.

Day 9 Practice

Before moving to Day 10, try answering these without looking:

1. What is routing?
2. What is a routing table?
3. What is a directly connected route?
4. What is a next hop?
5. What is a default route?
6. What does `0.0.0.0/0` mean?
7. What is longest-prefix matching?
8. Why does a router change the Ethernet MAC addresses?
9. Does the destination IP normally change when a packet passes through a router?
10. What is the difference between a switch's MAC table and a router's routing table?
11. What does `ip route show`?
12. What does `ip route get 8.8.8.8` help you determine?

Once these are clear, Day 10 will be ICMP → ping → traceroute , where we'll use the routing concepts from Days 7–9 to understand exactly how ping and traceroute work packet-by-packet.